Project Development Plan

Public Library Management System

1. Introduction

This document presents the phased development plan for the **Public Library Management System**. The project follows an industry-standard, backend-first, API-driven development approach to ensure data integrity, scalability, and long-term maintainability.

Each phase clearly specifies its objectives, tasks, deliverables, and estimated time duration.

2. Phase 0: Planning and Documentation

Estimated Time: 0.5 – 1 Day

Objectives

- Define project scope clearly
- Prevent feature creep
- Establish a shared understanding before implementation

Tasks

- Prepare initial documentation (README, architecture, API plan)
- Define user roles and permissions
- Finalize in-scope and out-of-scope features

Deliverables

- Approved project scope
- System architecture overview
- API contract document

3. Phase 1: Backend Foundation

Estimated Time: 2 Days

Objectives

- Establish backend as the single source of truth
- Implement core data models and APIs

Tasks

- Set up Node.js and Express server
- Configure MongoDB connection
- Implement Book and Admin schemas
- Develop CRUD APIs for books
- Test APIs using Postman

Deliverables

- Stable backend server
- Functional CRUD APIs
- Verified database operations

4. Phase 2: Inventory Business Logic

Estimated Time: 1 Day

Objectives

- Ensure correct inventory behavior
- Prevent inconsistent stock updates

Tasks

- Implement issue book functionality
- Implement return book functionality
- Enforce atomic availability updates
- Handle edge cases such as zero availability

Deliverables

- Reliable inventory logic
- Accurate available copy tracking

5. Phase 3: Real-Time Communication

Estimated Time: 1 – 1.5 Days

Objectives

- Enable real-time availability updates
- Eliminate manual page refresh requirements

Tasks

- Integrate Socket.IO on backend
- Emit real-time events on inventory changes
- Define event payload structure
- Test real-time communication using logs

Deliverables

- Functional real-time system
- Verified socket event emissions

6. Phase 4: Frontend Core Development

Estimated Time: 2 Days

Objectives

- Build public-facing user interface
- Consume backend APIs efficiently

Tasks

- Set up React application
- Develop catalog and book detail pages
- Build reusable UI components
- Fetch and display data using REST APIs

Deliverables

- Functional public catalog interface
- Accurate data rendering

7. Phase 5: Frontend Real-Time Integration

Estimated Time: 1 Day

Objectives

- Reflect inventory updates instantly on the UI

Tasks

- Integrate Socket.IO client
- Listen for availability update events
- Update UI state dynamically
- Handle socket reconnection scenarios

Deliverables

- Real-time UI updates
- Seamless user experience

8. Phase 6: Admin Dashboard

Estimated Time: 1 – 1.5 Days

Objectives

- Provide secure administrative controls

Tasks

- Implement admin authentication using JWT
- Build admin dashboard interface
- Add book management functionality
- Implement issue and return actions

Deliverables

- Secure admin interface
- Complete inventory management capabilities

9. Phase 7: Testing, Polish, and Finalization

Estimated Time: 1 Day

Objectives

- Prepare the project for evaluation and demonstration

Tasks

- Add loading and error states
- Improve UI consistency
- Fix bugs and handle edge cases
- Update project documentation

Deliverables

- Stable, demo-ready application
- Updated documentation

10. Overall Timeline Summary

Phase	Estimated Time
Planning	0.5 – 1 Day
Backend Foundation	2 Days
Inventory Logic	1 Day
Real-Time Integration	1 – 1.5 Days
Frontend Core	2 Days
Frontend Real-Time	1 Day
Admin Dashboard	1 – 1.5 Days
Testing & Polish	1 Day
Total Duration	**10 – 12 Days**

11. Conclusion

This phased development plan ensures a structured workflow, minimizes rework, and aligns with professional software engineering practices. Completing each phase fully before progressing ensures system stability, clarity, and production readiness.